

Definitions-Only Skeleton

Elementary Notation Conventions

Throughout, n is a positive integer.

The notation $[n]$ denotes the finite set $\{1, 2, \dots, n\}$.

A uniform random function $f : [n] \rightarrow [n]$ means a function chosen uniformly from the finite set of all functions from $[n]$ to $[n]$.

Source Definitions

Let $f : [n] \rightarrow [n]$, and let G_f be its associated directed graph, with vertex set $[n]$ and directed edges $(i, f(i))$ for $i \in [n]$. A vertex is *cyclic* if it belongs to a cycle of G_f .

Target Statement

Theorem. *The probability that a uniform random function $f : [n] \rightarrow [n]$ has a unique cyclic vertex is $\frac{1}{n}$.*

Source Statements Not Granted as Support

The following source statements are retained only as non-proof mathematical statements from the paper. They are *excluded* under the run's definitions-only support policy.

- Functions f having a unique cyclic vertex are in bijection with rooted trees on $[n]$ (the unique cyclic vertex is the root, and edges are oriented towards the root).
- Cayley's formula asserts that there are n^{n-2} trees on the vertex set $[n]$, or equivalently there are n^{n-1} rooted trees.
- Let H_n denote the height of a uniform random vertex in a uniform random rooted tree with n vertices. Then $1 + H_n$ has the same law as the time of first repetition in a i.i.d. sequence of uniform variables on $[n]$ (time of first collision in the coupon collector process).